

Terna Engineering College

Computer Engineering Department

Program: Sem VI

Course: Cloud Computing Lab (CSL603)

Faculty: PREETI PATIL

Experiment No.4

A.1 Aim:

To study and Implement Platform as a Service using AWS Elastic Beanstalk/ Microsoft Azure App Service.

A.2 Prerequisite:

Understanding of Virtualization, Basics of Networking, Basics of security and privacy.

A.3 Objective:

To demonstrate the steps to deploy Web applications or Web services written in different languages on AWS Elastic Beanstalk/ Microsoft Azure App Service.

A.3 Outcome: (LO2)

After successful completion of this experiment students will be able to deploy the web application using AWS Elastic Beanstalk.

A.4 Theory:

AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with popular programming languages such as Java, .NET, PHP, Node.js, Python and Ruby. You simply upload your application and Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling and application health monitoring. At the same time, with Elastic Beanstalk, you retain full control over the AWS resources powering your application and can access the underlying resources at any time.

Most existing application containers or platform-as-a-service solutions, while reducing the amount of programming required, significantly diminish developers' flexibility and control. Developers are forced to live with all the decisions pre-determined by the vendor - with little to no opportunity to take back control over various parts of their application's infrastructure. However, with Elastic Beanstalk, you retain full control over the AWS resources powering your application. If you decide you want to take over some (or all) of

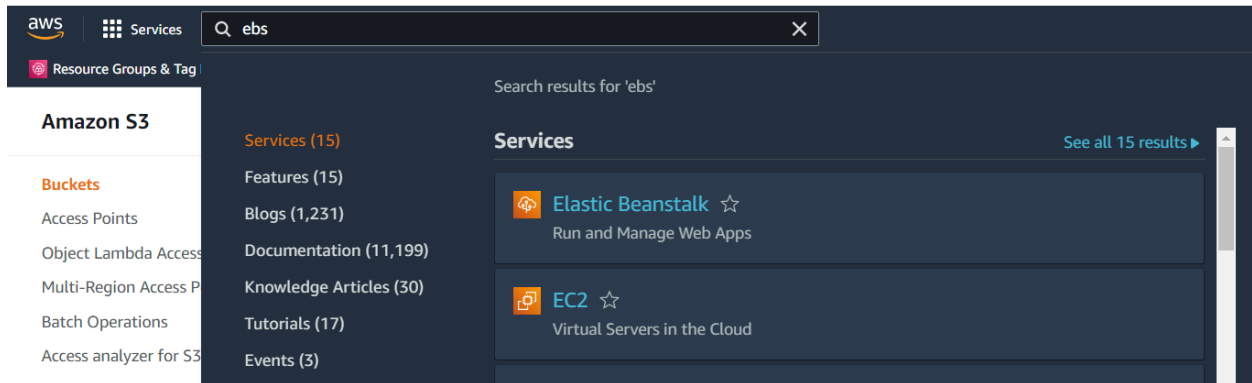
the elements of their infrastructure, you can do so seamlessly by using Elastic Beanstalk's management capabilities.

To ensure easy portability of your application, Elastic Beanstalk is built using familiar application/web servers such as Apache HTTP Server, Apache Tomcat, Nginx, Passenger and IIS 7.5/8.

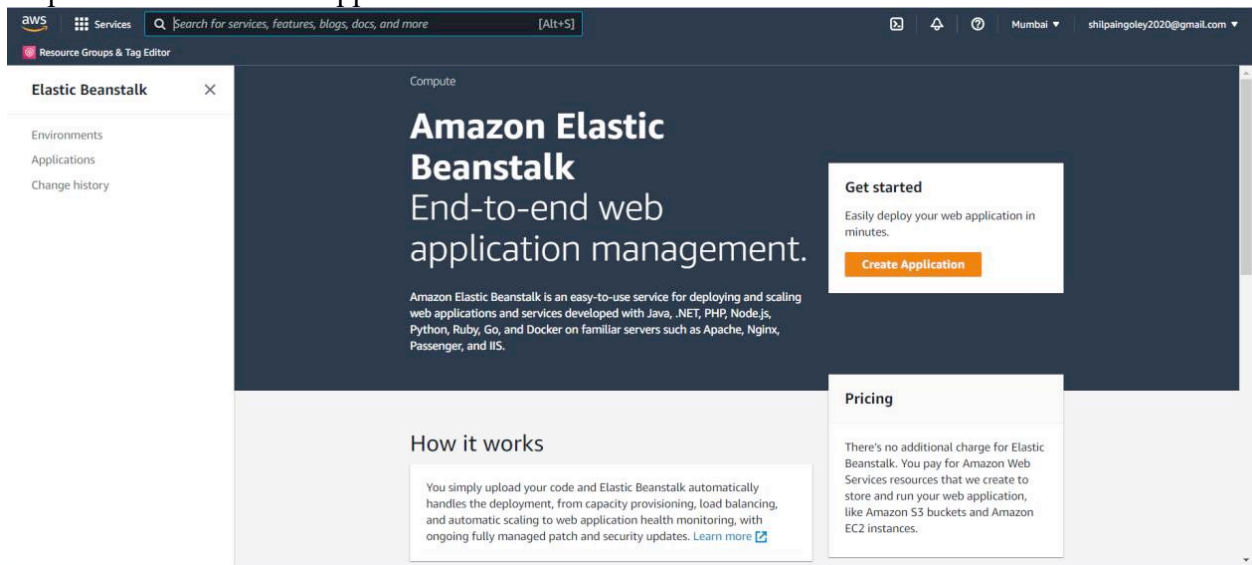
Following are steps to host a website using Elastic Beanstalk:

(Creating Sample Application)

Step1 : Login to AWS console and go to Elastic Beanstalk



Step 2: Click on Create Application



Step 3: Write Application information: Name, Tag,Platform etc.

aws Services Search for services, features, blogs, docs, and more [Alt+S] Mumbai

Resource Groups & Tag Editor

Elastic Beanstalk ✕

- Environments
- Applications
- Change history

Application information

Application name

Mynewappshilpa

Up to 100 Unicode characters, not including forward slash (/).

Application tags

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

Key	Value	
name	selfWebSite	Remove tag

Add tag

49 remaining

Step 4: In Application Code: select sample application and then Click on button Create Application

aws Services Search for services, features, blogs, docs, and more [Alt+S] Mumbai

Resource Groups & Tag Editor

Elastic Beanstalk ✕

- Environments
- Applications
- Change history

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case-sensitive. [Learn more](#)

Key	Value	
name	selfWebSite	Remove tag

Add tag

49 remaining

Platform

Platform

PHP

Platform branch

PHP 8.0 running on 64bit Amazon Linux 2

Platform version

3.3.10 (Recommended)

This will take a few minutes.

Step 5: Click on Environments -> Check the health of Environment wait till it becomes 'OK'

Application code

☒ Sample application
Get started right away with sample code.

☐ Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.

Cancel

Configure more options

Create application

AWS

Services

Search for services, features, blogs, docs, and more

[Alt+S]

Resource Groups & Tag Editor

Environments

Applications

Change history

▼ apptestshilpa

Application versions

Saved configurations

▶ Apptestshilpa-env

Elastic Beanstalk > Environments > Apptestshilpa-env

Creating Apptestshilpa-env

This will take a few minutes .

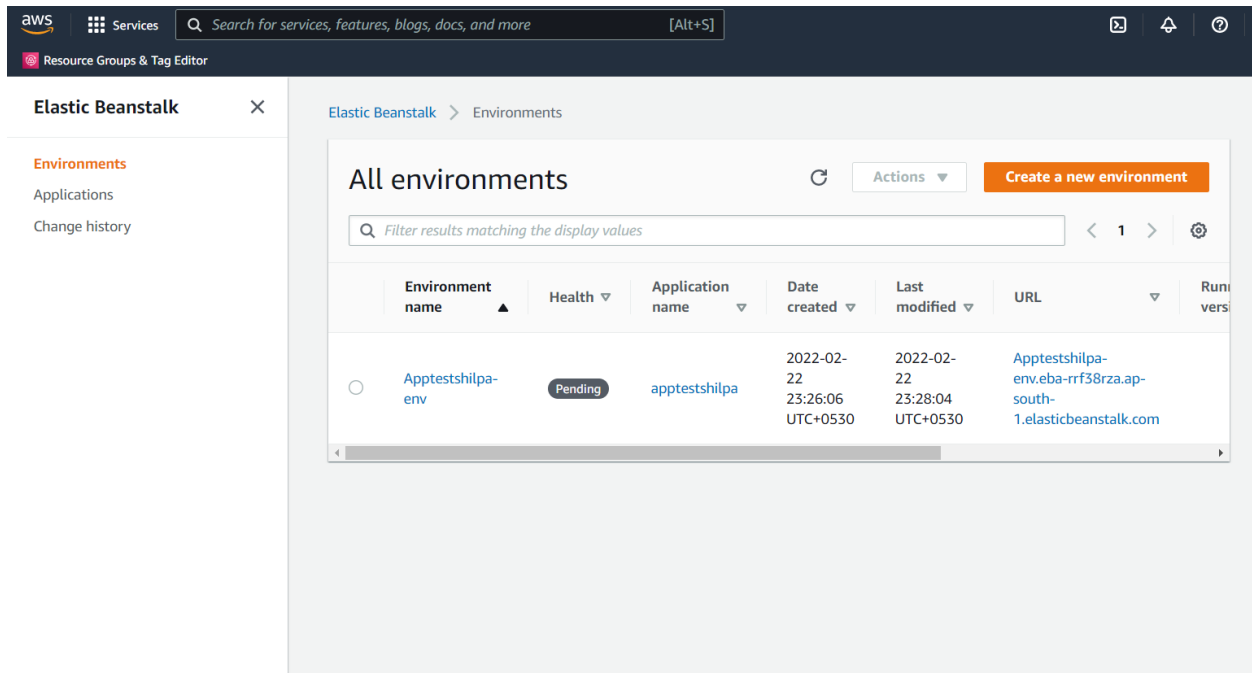
11:26pm Created security group named:
sg-0eb4e480238fb8826

11:26pm Created target group named:
arn:aws:elasticloadbalancing:ap-south-1:580032287469:targetgroup/awseb-AWSEB-ZN1T36EQ9LKB/790b52a1d8640215

11:26pm Using elasticbeanstalk-ap-south-1-580032287469 as Amazon S3 storage bucket for environment data.

11:26pm createEnvironment is starting.

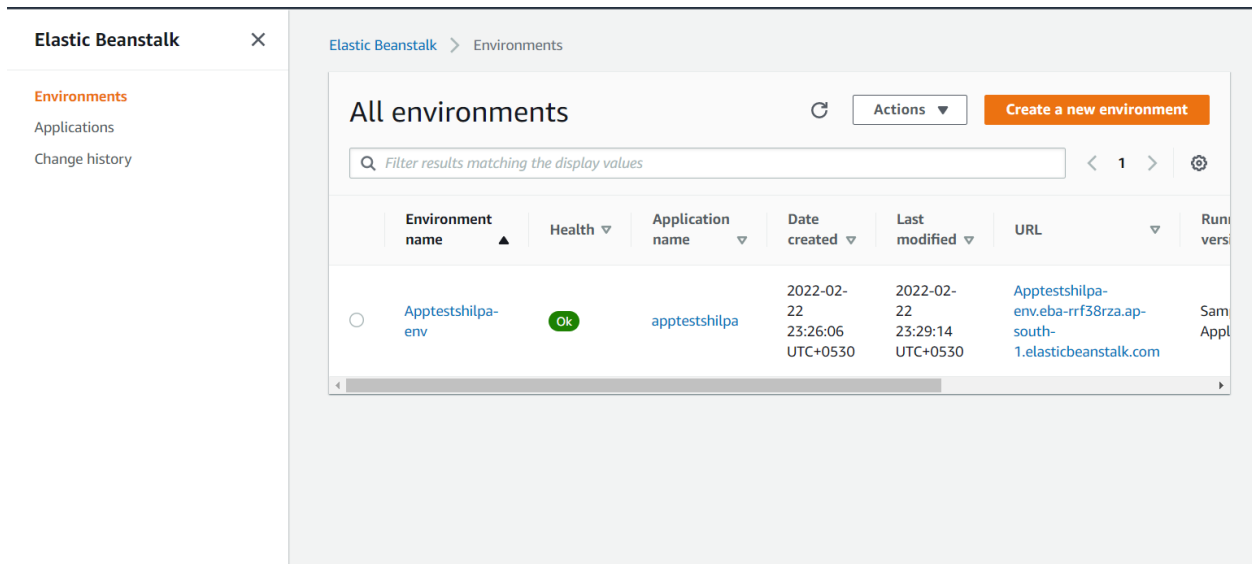
Step 6: Click the URL



The screenshot shows the AWS Elastic Beanstalk console. The left sidebar has 'Elastic Beanstalk' selected. The main content area is titled 'All environments'. A table lists the environments:

Environment name	Health	Application name	Date created	Last modified	URL	Run version
Apptestshilpa-env	Pending	apptestshilpa	2022-02-22 23:26:06 UTC+0530	2022-02-22 23:28:04 UTC+0530	Apptestshilpa-env.eba-rrf38rza.ap-south-1.elasticbeanstalk.com	

To Delete the application and Environment (Select it and in **Action** -Delete/Terminate : give conformation)



The screenshot shows the AWS Elastic Beanstalk console. The left sidebar has 'Elastic Beanstalk' selected. The main content area is titled 'All environments'. A table lists the environments:

Environment name	Health	Application name	Date created	Last modified	URL	Run version
Apptestshilpa-env	Ok	apptestshilpa	2022-02-22 23:26:06 UTC+0530	2022-02-22 23:29:14 UTC+0530	Apptestshilpa-env.eba-rrf38rza.ap-south-1.elasticbeanstalk.com	Sam Appl

Congratulations!

Your AWS Elastic Beanstalk *PHP* application is now running on your own dedicated environment in the AWS Cloud

You are running PHP version 8.0.13

This environment is launched with Elastic Beanstalk PHP Platform

What's Next?

- [AWS Elastic Beanstalk overview](#)
- [Deploying AWS Elastic Beanstalk Applications in PHP Using Eb and Git](#)
- [Using Amazon RDS with PHP](#)
- [Customizing the Software on EC2 Instances](#)
- [Customizing Environment Resources](#)

AWS SDK for PHP

- [AWS SDK for PHP home](#)
- [PHP developer center](#)
- [AWS SDK for PHP on GitHub](#)

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case there is no ERP access available)

Roll No. C54	Name:Quazi Md Moinuddin
Class :TE-C Comps	Batch :C3
Date of Experiment:	Date of Submission:5/2/25
Grade :	

B.1 Question of Curiosity:

(To be answered by student based on the practical performed and learning/observations)

Q1: What are the benefits of Paas in cloud computing?

Platform as a Service (PaaS) in cloud computing offers several benefits, including:

- 1 **Simplicity:** PaaS provides a complete development environment, reducing complexity by allowing developers to focus on building applications without worrying about underlying infrastructure.
- 2 **Scalability:** PaaS solutions can automatically scale resources up or down based on demand, ensuring efficient use of resources and cost savings.
- 3 **Cost-effective:** It eliminates the need for purchasing and maintaining hardware and software, thus reducing upfront costs.
- 4 **Flexibility:** PaaS supports multiple programming languages and tools, allowing developers to work with technologies they are familiar with.
- 5 **Collaboration:** It facilitates team collaboration by enabling developers to work on projects from different locations simultaneously.

Integrated Tools: PaaS services often come with integrated tools for database management, development, testing, and deployment, streamlining the application lifecycle.

Q2:What is Paas?

Ans : PaaS is a cloud computing service that uses virtualization to offer an application development platform to developers or organizations. This platform includes computing, memory, storage, database and other app development services. PaaS solutions can be used to develop software for internal use or offered for sale.

PaaS technology offers a company virtual infrastructure, such as data centers, servers, storage and network equipment, plus an intermediate layer of software, which includes tools for building apps. Of course, a user interface is also part of the package to provide usability.

Customers can deploy PaaS in one of three different cloud deployment models defined by the National Institute of Standards Technology (NIST) as follows:

- **Private Cloud:** The development platform is built on infrastructure that is provisioned for exclusive use by a single organization comprising multiple consumers. The infrastructure may be owned, managed and operated by the organization, a third party or some combination, and it may exist on or off premises.
- **Public Cloud:** The development platform is built on infrastructure provisioned for use by multiple organizations (also known as a multi-tenant model). The infrastructure may be owned, managed and operated by a business, academic or government organization, or some combination. It exists on the premises of the cloud provider.
- **Hybrid Cloud:** The development platform is built across both public cloud and private cloud. The two cloud models remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability. Hybrid cloud is rarely used for PaaS solutions.

Q3) what is elastic bean stack ?

Ans : AWS Elastic Beanstalk is an AWS managed service for web applications. Elastic beanstalk is a preconfigured EC2 server that can directly take up your application code and environment configurations and use it to automatically provision and deploy the required resources within AWS to run the web application. Unlike EC-2 which is Infrastructure as a service, Elastic beanstalk is a Platform As A service (PAAS) as it allows users to directly use a pre-configured server for their application. Of course you can deploy applications without ever having to use elastic beanstalk but that would mean having to choose the appropriate service from the vast array of services offered by AWS, manually provisioning these AWS resources and stitching them up together to form a complete web-application. Elastic beanstalk abstracts the underlying configuration work and allows you as a user to focus on more pressing matters.

This raises a concern that if elastic beanstalk configures most of the resources itself then it abstracts the underlying details. Can developers change configuration if needed? The answer is Yes. Elastic Beanstalk is provided to make application deployment simpler but at no level will it restrict the developers from changing any configurations. AWS Elastic Beanstalk Features:

- **Application:** Elastic Beanstalk directly takes in our project code. So Elastic Beanstalk application is named the same as your project home directory.
- **Application Environments:** Users may want their application to run on different environments like DEV, UAT and PROD. You can create and configure different environments to run application on different stages.
- **Environment Health:** One of the most lucrative features about running application on AWS or most of the other cloud platforms is the automated health checks. AWS runs automatic health checks on all EC-2 deployments (Elastic Beanstalk is a managed EC-2 service) which can be monitored from AWS console. For example, in case of web applications AWS will regularly, as scheduled by the developers, ping the application to check if the response is status code 200 and the application is running as expected. Health check responses:

- Red: Application failed all health tests.
- Yellow: Application failed some of the health tests.
- Grey: Application is updating.
- Green: Application passed health check successfully.
- Isolated: All environments within a single application are isolated from each other (independent of each others' running states). Needless to say two different applications are also isolated.
- Scalability: Using Auto-Scaling within Elastic beanstalk makes the application dynamically scalable.
- Elastic Load Balancing: All the web requests to the application are not directly relayed to application instances. They first hit the Elastic Load Balancer (ELB), which, as the name suggests, balances the load across all the application instances.
- Language support: Elastic Beanstalk supports the applications developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker on familiar servers such as Apache, Nginx, Passenger, and IIS.
- Pricing: There is no extra charge for using Elastic Beanstalk. Users are only required to pay for the services and resources provisioned by Elastic Beanstalk Service.
- Automatic Provisioning: Elastic Beanstalk takes away the burden of choosing the right services and configuring their security groups to work together.
- Impossible to Outgrow: AWS claims that since Elastic Beanstalk uses Auto Scaling feature it can, in theory, handle any amount of internet traffic

B.2 Conclusion:

(Students must write the conclusion as per the attainment of individual outcome listed above and learning/observation)

Step 2
Configure service access

Step 3 - optional
Set up networking, database, and tags

Step 4 - optional
Configure instance traffic and scaling

Step 5 - optional
Configure updates, monitoring, and logging

Step 6
Review

Environment tier [Info](#)

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

☒ Web server environment
Run a website, web application, or web API that serves HTTP requests. [Learn more](#)

☐ Worker environment
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. [Learn more](#)

Application information [Info](#)

Application name

Maximum length of 100 characters.

▶ Application tags (optional)

Environment information [Info](#)

Choose the name, subdomain and description for your environment. These cannot be changed later.

Environment name

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Domain
.ap-south-1.elasticbeanstalk.com

Environment description

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Platform [Info](#)

Platform type

☒ Managed platform
Platforms published and maintained by Amazon Elastic Beanstalk. [Learn more](#)

☐ Custom platform
Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform

Platform branch

Platform version

Application code [Info](#)

☒ Sample application

☐ Existing version
Application versions that you have uploaded.

☐ Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.

Presets [Info](#)

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Configuration presets

☒ Single instance (free tier eligible)

☐ Single instance (using spot instance)

☐ High availability

☐ High availability (using spot and on-demand instances)

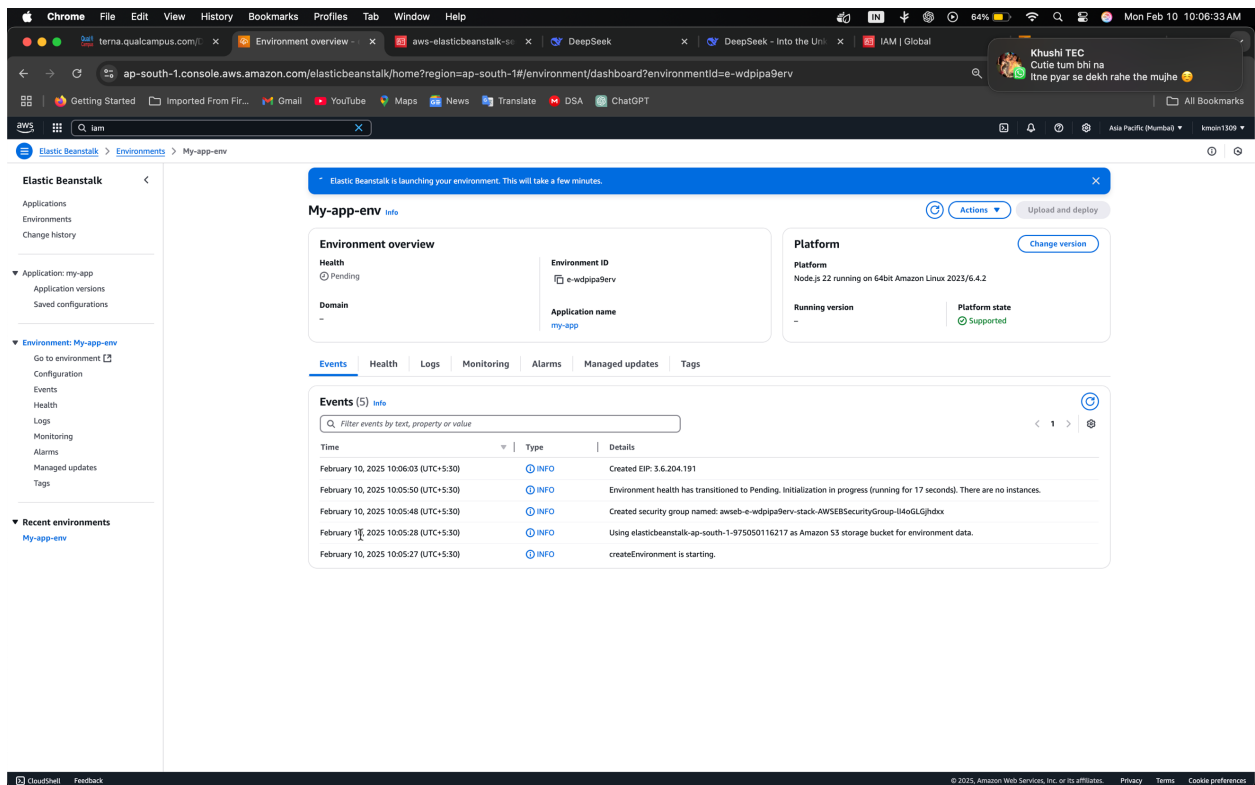
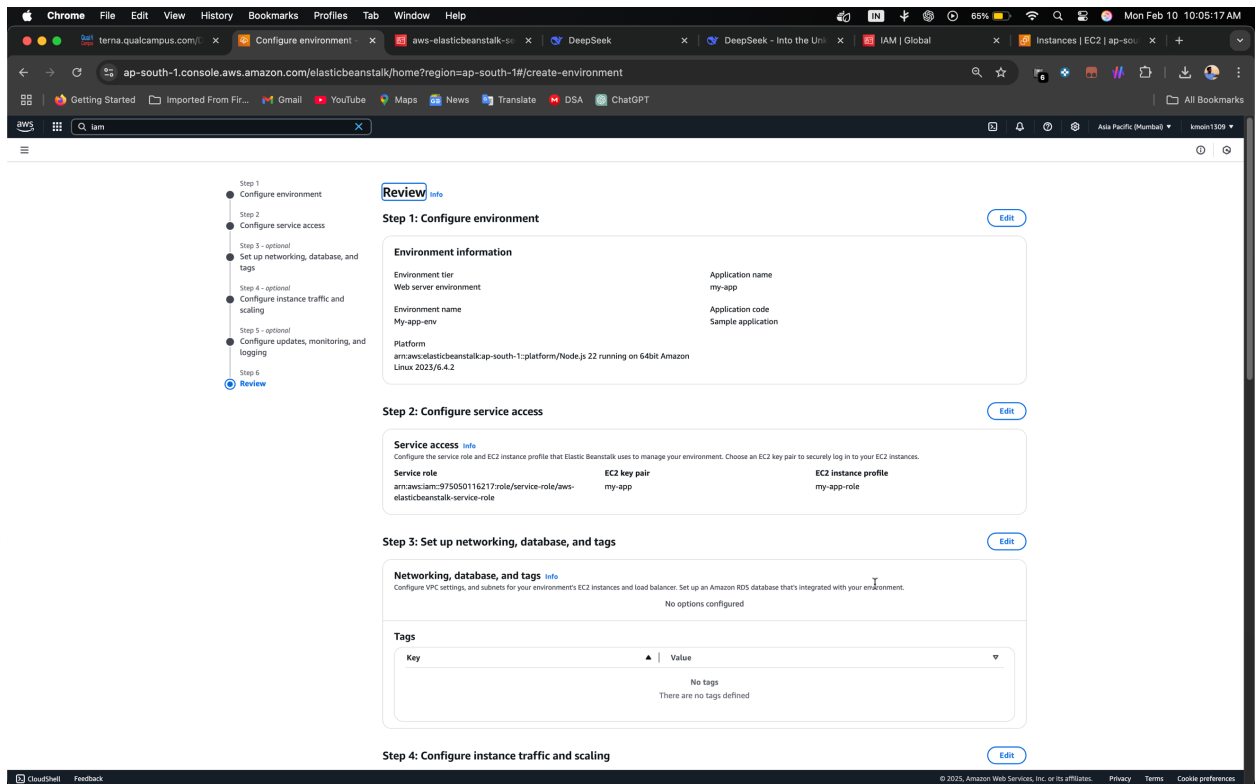
☐ Custom configuration

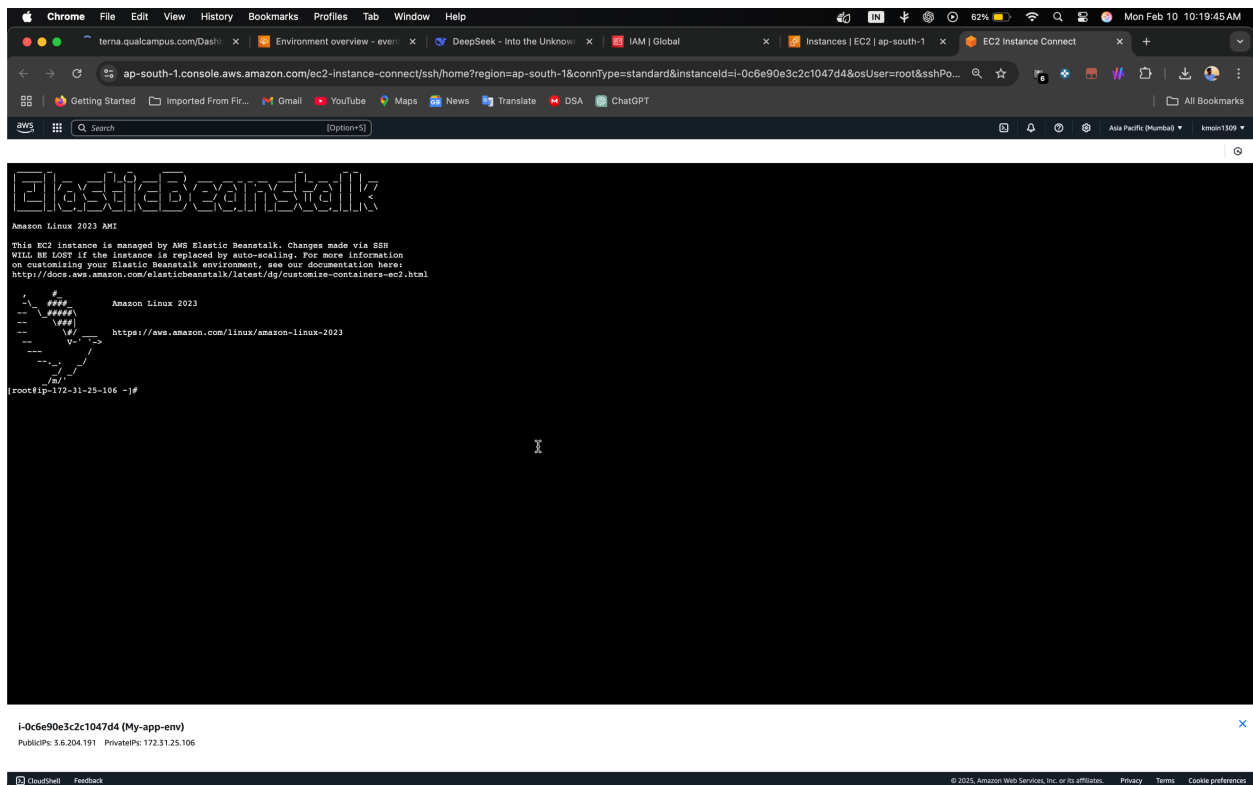
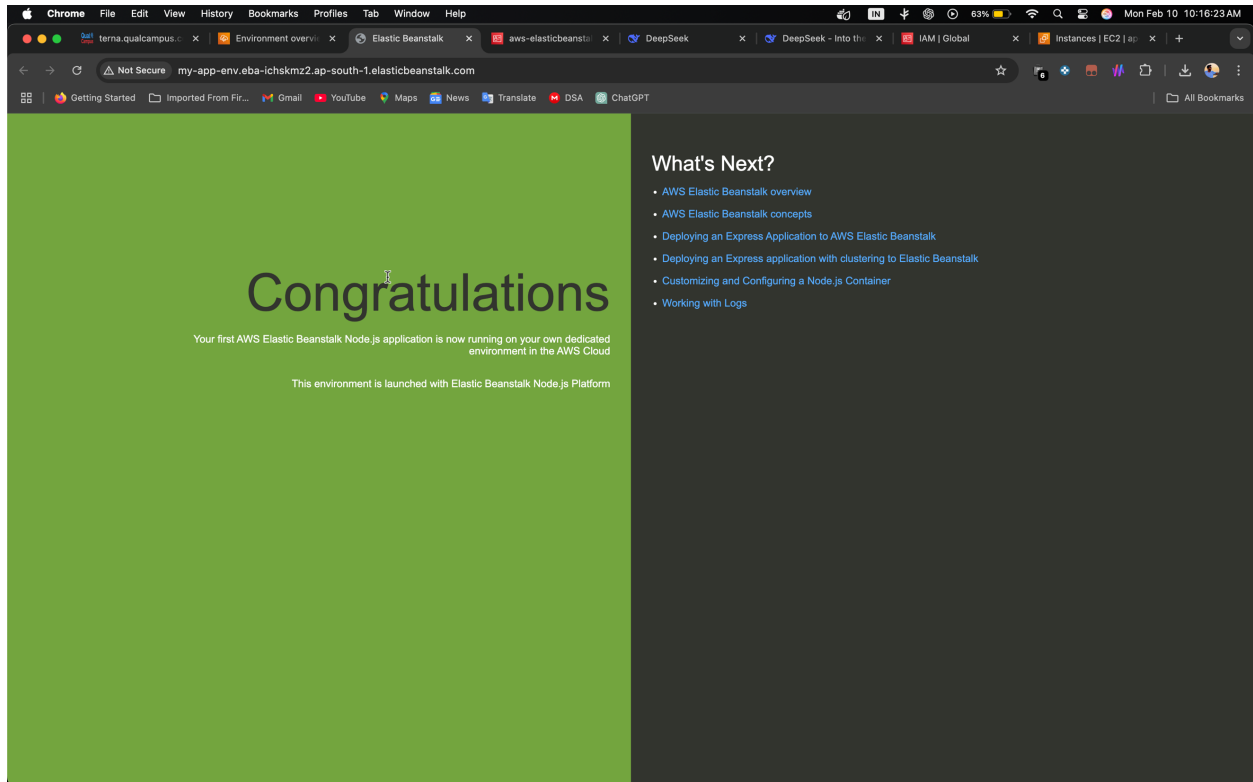
Cancel

Next

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)





In conclusion, the practical deployment of a web application using AWS Elastic Beanstalk demonstrated the platform's user-friendly approach, automatic scaling, and support for diverse programming languages. Students gained hands-on experience in deploying, monitoring, and managing applications in a Platform as a Service (PaaS) environment.